

Chapter 37: Orthogonality Worksheet

1. Are each of the following statements true or false?

(a) If $\{\mathbf{v}_1, \mathbf{v}_2\}$ is an orthogonal basis, then $\{r_1\mathbf{v}_1, r_2\mathbf{v}_2\}$ is also an orthogonal basis for any r_1, r_2 with $r_1 \neq 0, r_2 \neq 0$.

(b) For the inner product on $\mathbb{R}^2$ defined by $\langle \mathbf{u}, \mathbf{v} \rangle = u_1v_1 + 5u_2v_2$, the vectors $\mathbf{u} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ are orthogonal.

(c) If $\mathbf{u}, \mathbf{v}$ are orthogonal, then $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

(d) Let V be an inner product space. If $\mathbf{v}$ is orthogonal to $\mathbf{w}_1, \dots, \mathbf{w}_n$, then $\mathbf{v}$ is orthogonal to every vector in the span of $\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$.

2. Consider P_2 with the inner product $\langle f, g \rangle = \int_0^1 f(x)g(x)dx$. Consider the subspace W of P_2 of all polynomials orthogonal to $p(x) = 1 - 2x$. Find a basis for W .

3. Let $B = (\mathbf{b}_1, \mathbf{b}_2)$ be an orthonormal ordered basis for $\mathbb{R}^2$ (with respect to the standard inner product) where

$$\mathbf{b}_1 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}, \mathbf{b}_2 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} \end{bmatrix}.$$

Find the coordinate vector for $\mathbf{x}$ with respect to B , where $\mathbf{x} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$.

4. Verify that the following set

$$S = \left\{ \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \right\}$$

is an orthogonal basis for $\mathbb{R}^3$ (with respect to the dot product), then find an orthonormal basis using S .